Chair Balancing Act

Continuous audio escalation for tilting chairs

All legs down = silence. Any leg lifts = rising tone.

Part safety reminder, part office prank, part gift-shop impulse buy.

How It Works

Static Balance

SILENCE

- All legs on the floor
- Device sleeps, no sound
- MCU in deep sleep (<1 uA)

Dynamic Balance

CONTINUOUS RISING TONE

- Any leg lifts off the floor
- Pleasant low tone begins
- Rises smoothly with tilt angle

Near Fall

MAXIMUM ALARM

- Steep angle, near tipping
- Tone at peak urgency
- Everyone knows what's happening

32mm round x 10mm | 15g | CR2450 coin cell | Adhesive mount

Fixed-leg chairs only (school, dining)

Architecture

Data Flow

Accelerometer --> MCU --> Class-D Amp --> Piezo

I2C + interrupt PWM tone gen sound out

SUBSYSTEMS

Accelerometer (LIS2DH12)

3-axis tilt, wake interrupt, I2C

MCU (ATtiny1616)

Filter, tone gen (PWM), calibration, sleep

Class-D Amp (PAM8302)

Amplifies MCU PWM to drive piezo

Piezo Speaker (13mm)

75-85 dB at 10cm, fits 10mm enclosure

CR2450 Coin Cell

600 mAh, months of life, user-replaceable

Button (1x tactile)

Mode cycle + calibration + on/off

No app. No cloud. No BLE. No WiFi. No clips.

MCU generates tone in real-time. 60 seconds to first use.

FIRMWARE

Bare-metal (no RTOS) on ATtiny1616 (16KB flash, 2KB SRAM)
8 modules: sleep mgr, tilt reader, filter, tone gen,
escalation curves, mode mgr, calibration, battery monitor
Boot-to-ready: < 3 seconds. Tone generated via HW PWM.

Constraints & BOM

BOM cost — **< $8 at 1k units**

Retail $15-25, gift/novelty price

Battery life — **> 6 months (CR2450)**

20 tilt events/day, ~9 months typical

Latency — **< 500ms tilt-to-sound**

Comedy timing is the product

Size — **35mm dia x 10mm**

Hidden under chair seat

Weight — **< 18g assembled**

Adhesive pad holds it up

Drop survival — **1.5m onto hard floor**

Falls off chairs regularly

BOM ESTIMATE (1k units)

ATtiny1616 MCU	$0.60
LIS2DH12 accelerometer	$0.80
Class-D amp (PAM8302)	$0.30
Piezo disc (13mm)	$0.30
CR2450 holder + battery	$0.65
PCB (30x25mm, 2-layer)	$0.50
Passives + button	$0.25
Enclosure (ABS molded)	$1.50
Adhesive pad + packaging	$1.00
Assembly + test	$1.50
Total COGS	**~$7.40**

Hardest Problems

1 Escalation curve across chair types

School chairs and dining chairs have different fall points and tilt ranges. The continuous tone must start gently at first leg lift and peak just before the fall point -- not too early (annoying) or too late (useless). Needs auto-calibration + playtesting on 5+ types.

2 Filtering fidgeting from tilting

Sitting down, crossing legs, leaning to grab something all produce accelerometer signals. The continuous model means any sustained tilt produces sound -- the dead-band (~2 deg) and 300ms time filter are critical to avoid false alarms.

3 Designing the continuous tone

The tilt-to-tone mapping IS the product. Must be agreeable at low tilt and urgent near the fall point. Three modes need distinct character. Needs sound designer + playtesting with kids.

Hardware is simple. The product lives or dies on the escalation curve -- how tilt maps to tone. Playtesting > engineering.

PRD: 45 Requirements

GATE: PASS

66 pass / 23 N/A / 1 gap

BY TRACK

Hardware — 18

8 electrical + 8 mechanical + 4 PCB

Firmware — 18

7 core + 4 modes + 4 calibration + 2 power + 1 versioning

Integration — 9

End-to-end tests spanning HW + FW

BY PRIORITY

40 Must-have **5 Should-have**

0 Nice-to-have (deferred to V2)

KEY REQUIREMENTS

FW-C-05	Continuous tone scales with tilt angle
FW-C-06	Angle-to-tone tracking < 100ms
INT-02	Zero false triggers on transient motion
FW-PM-04	Battery life >= 6 months at 20 events/day
HW-M-01	Enclosure max 35mm dia x 10mm height
HW-E-06	Total sleep current <= 2 uA

TRACEABILITY

S1 Dinner table -> FW-C-01..07, INT-01, INT-03
S2 Classroom -> FW-M-01, FW-M-04, INT-04, HW-M-06
S3 First use -> FW-CAL-01, FW-CAL-04, HW-M-06
S4 Battery swap -> HW-M-04, FW-PM-02..03, INT-06
S5 False trigger -> FW-C-03, FW-C-04, INT-02

Open Items & V2

7 OPEN ITEMS (resolve before production)

M2	Escalation curve tuning across 5+ chair types
M2	Piezo volume validation in real environments
M2	False trigger testing with real users
M2	PWM audio quality -- pleasant at low tilt?
M2	CPSIA / EN 71 acoustic safety check
M3	Tone design for 3 modes (sound designer + FW)
M3	Adhesive durability testing

V2 HORIZON (if market validates)

Custom tone profiles via USB

Board respin: USB-C, upload escalation curve params

BLE + companion app

MCU swap: ATtiny -> nRF52. Full board respin + app dev

Adjustable sensitivity

Requires BLE. Depends on companion app decision

Rechargeable battery

Adds charging IC, USB-C, Li-Po. Enclosure redesign

V2 decisions wait for 6+ months of V1 sales data and customer feedback.